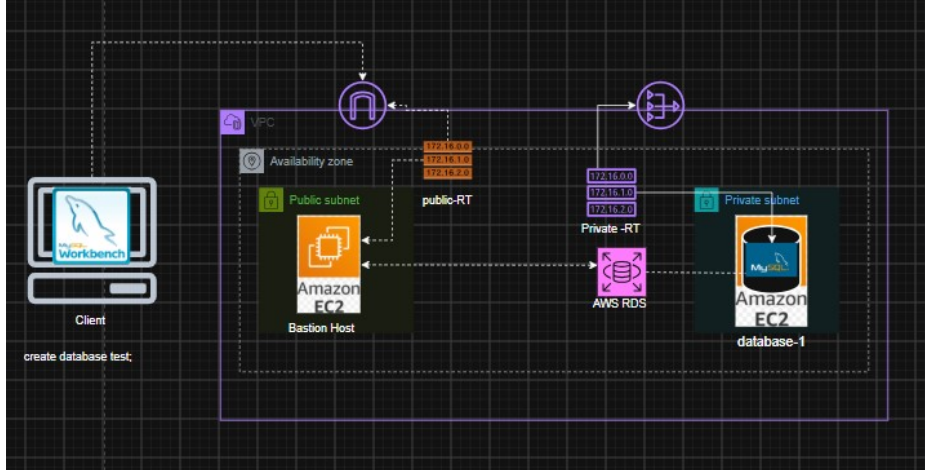


Access RDS DB deployed in Private server using MySQL Workbench



Task -1

Deploy a database using AWS RDS on a private server and communicate with database using MySQL Workbench. (Here we are using free tier, so will be using Single-AZ DB instance deployment (1 instance), we can also use Multi-AZ DB cluster deployment (3 instances), and Multi-AZ DB instance deployment (2 instances))

Prerequisite :-

1. VPC and More (1 az ,1 public ,1 private subnet , NAT – required, PVC endpoint – not required).
2. create one database In created vpc by selecting private subnet (access type private) .
3. Create a Bastion host in public subnet(with private key , SG-all traffic ,anywhere).
4. Configure MySQL workbench (MySQL workbench -> Bastion host->private database).

Steps –

1. Create vpc .

VPC > Your VPCs > Create VPC

Encryption settings - optional

Number of Availability Zones (AZs) info
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
1 2 3

Customize AZs

Number of public subnets info
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.
0 **1**

Number of private subnets info
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.
0 **1** 2

Customize subnets CIDR blocks

NAT gateways (5) updated info
NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.
None **Regional - new** Zenal

Introducing regional NAT gateway
AWS now offers a multi-AZ NAT Gateway, eliminating the need for separate NAT Gateways across availability zones.

VPC endpoints info
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.
None S3 Gateway

Preview

VPC Show details
Your AWS virtual network
MyVPC-vpc

Subnets (2)
Subnets within this VPC
us-east-1a
MyVPC-subnet-public-1-us-east-1a
MyVPC-subnet-private-1-us-east-1a

Route tables (3)
Route network traffic to resources
MyVPC-rtb-public
MyVPC-rtb-private-1-us-east-1a
rtb-regional-nat

Your VPCs

VPCs VPC encryption controls

Your VPCs (5) info

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option
☐	MyVPC-vpc	vpc-045aa353692240ad7	Available	-	-	Off	10.0.0.0/16	-	dopt-01a1d5f
☐	-	vpc-04db7c337e8062165	Available	-	-	Off	172.30.0.0/16	-	dopt-01a1d5f

2. Create database , Select - Single-AZ DB instance deployment (1 instance).

DB instant identifier – database-1

Master user name – admin

Credential management – self managed

Instance type – t3micro

Create database info

Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type info

☒ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☒ MySQL

☐ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates
Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

☒ Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

- Chose VPC
- Chose subnet group (We need to create subnet group to indicate where our replica we want RDS to deploy)

Create DB subnet group
To create a new subnet group, give it a name and a description, and choose an existing VPC. You will then be able to add subnets related to that VPC.

Subnet group details
You won't be able to modify the name after your subnet group has been created.
Name:
Must contain from 1 to 255 characters. Alphabetic characters, spaces, hyphens, underscores, and periods are allowed.
Description:
VPC: Choose a VPC identifier that corresponds to the subnets you want to use for your DB subnet group. You won't be able to choose a different VPC identifier after your subnet group has been created.
MyVPC-vpc (vpc-045aa35692240ad2)
3 Subnets, 1 Availability Zones

Add subnets
Availability Zones
Choose the Availability Zones that include the subnets you want to add.
Choose an availability zone:
Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.
Select subnets:
Subnet ID: subnet-056b078d5db90a7b CIDR: 10.0.128.0/20
For Multi-AZ DB clusters, you must select 1 subnets in 1 different Availability Zones.

Subnets selected (1)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	MyVPC-subnet-private1-us-east-1a	subnet-056b078d5db90a7b	10.0.128.0/20

Cancel Create

C.

Create database

Virtual private cloud (VPC)
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.
MyVPC-vpc (vpc-045aa35692240ad2)
3 Subnets, 2 Availability Zones
After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.
After a database is created, you can't change its VPC.

DB subnet group
Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.
myvpc-private-subnet-1b
2 Subnets, 2 Availability Zones

Public access
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
Choose existing
Choose existing VPC security groups
Create new
Create new VPC security group

Existing VPC security groups
Choose one or more options
default

Availability Zone
us-east-1a

Select SG – allow inbound for 3306 port for vpc connection .

Availability zone – select in which zone you want to deploy your db

VPC security group (firewall) [Info](#)
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing
Choose existing VPC security group

☐ Create new
Create new VPC security group

Existing VPC security groups
Choose one or more options

default

Availability Zone [Info](#)
us-east-1a

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)
Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rdc-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

Then create database. It will take some time

Databases

Successfully created database database-1
You can use settings from database-1 to simplify configuration of suggested database add-ons while we finish creating your DB for you. [View connection details](#)

Databases (1) [Group resources](#) [Modify](#) [Actions](#) [Create database](#)

Filter by databases

	DB identifier	Status	Role	Engine	Upgrade rollout order	Region	Size	Recommendations	CPU	Current activity
	database-1	Available	Instance	MySQL Co...	SECOND	us-east-1a	db.t3.micro		5.64%	0 Conn

3. Create a Bastion host in public subnet(with private key , SG-all traffic ,anywhere).

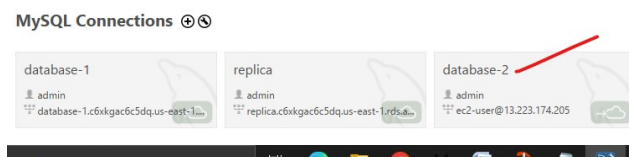
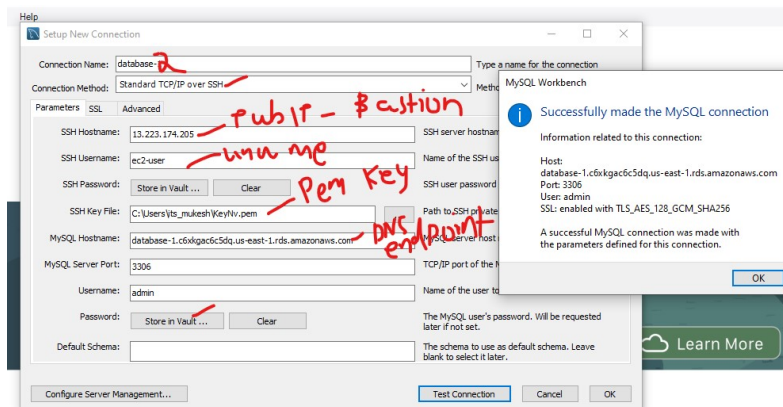
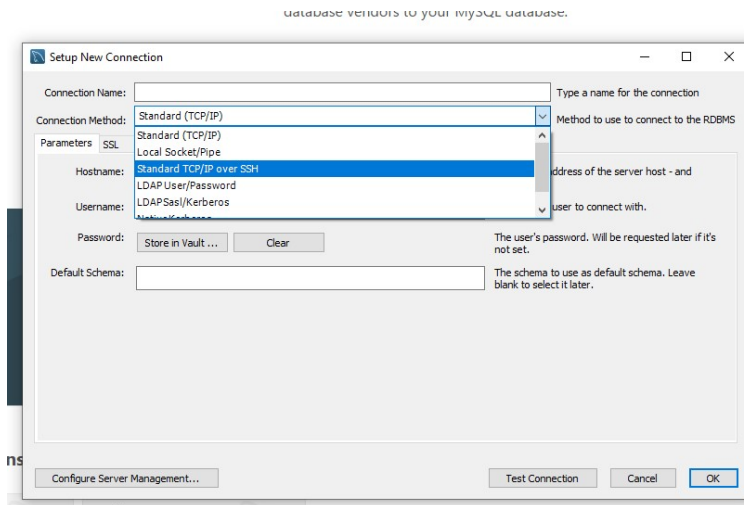
Instances > i-076fa517e9989ee6c

Instance summary for i-076fa517e9989ee6c (Bastion -host) [Info](#)
Updated less than a minute ago

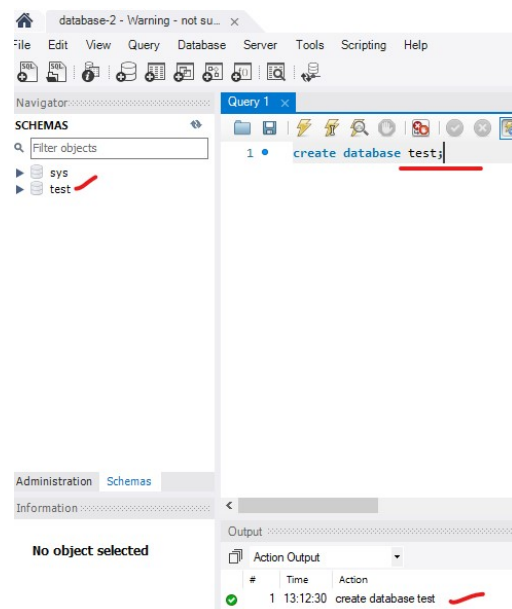
[Connect](#) [Instance state](#) [Actions](#)

Instance ID i-076fa517e9989ee6c	Public IPv4 address 13.223.174.205 open address	Private IPv4 addresses 10.0.9.245
IPv6 address -	Instance state Running	Public DNS ec2-13-223-174-205.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-9-245.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-9-245.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding It is taking a bit longer than usual to fetch your data
Auto-assigned IP address 13.223.174.205 [Public IP]	VPC ID vpc-045aa355692240ad2 (MyVPC-vpc) open address	Auto Scaling Group name -
IAM role -	Subnet ID subnet-0190268a830bdcda0 (MyVPC-subnet-public1-us-east-1a) open address	Managed False
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:198302589656:instance/i-076fa517e9989ee6c	
Operator -		

4. Configure MySQL workbench (MySQL workbench -> Bastion host->private database).
Select TCP/IP over SSH



Now make connection and create database name test –



Lab Over